Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

HHS Public Access
Author manuscript
Pediatr Blood Cancer. Author manuscript; available in PMC 2025 November 18.

Published in final edited form as:
Pediatr Blood Cancer. 2025 June ; 72(6): e31653. doi:10.1002/pbc.31653.

Risk for unplanned pregnancy among survivors of childhood cancer

Brooke Cherven, PhD[1,2], **Kayla Fitch, PsyD**[1,3], **Lauren Quast Esposito, PhD**[2], **Ebonee Harris, MPH**[1], **Karen Burns, MD**[4,5], **Jenna Demedis, MD, MS**[6,7], **Holly Hoefgen, MD**[8], **Ann Mertens, PhD**[1,2], **James L. Klosky, PhD, ABPP**[1,2]

[1]Aflac Cancer and Blood Disorders Center at Children's Healthcare of Atlanta

[2]Department of Pediatrics, Emory University School of Medicine

[3]Wellstar KenMar Pediatrics

[4]University of Cincinnati College of Medicine

[5]Cincinnati Children's Hospital Medical Center

[6]Center for Cancer and Blood Disorders at Children's Hospital Colorado

[7]Department of Pediatrics, University of Colorado School of Medicine

[8]Washington University School of Medicine, St. Louis, MO

Abstract

The purpose of this study was to identify factors associated with risk for unplanned pregnancy in sexually active female survivors using self-report survey data. Risk for unplanned pregnancy was defined as using less effective/no contraception while also not desiring pregnancy. Of N=160 participants (age 24.0±3.1y), 33.1% were at-risk for unplanned pregnancy. On multivariable analysis, participants were less likely to be categorized at-risk for unplanned pregnancy if they reported diagnosis of ovarian failure/premature menopause (OR 0.16, 95%CI 0.02–0.72, p=.032), greater concerns about fertility (OR 0.59, 95%CI 0.43–0.80, p=.001), and religious identity of agnostic/atheist (compared with Christian, OR 0.10, 95%CI 0.01–0.41, p=.005).

Keywords

childhood cancer survivor; unplanned pregnancy; contraception

There are nearly 500,000 of survivors of childhood cancer in the United States, many of whom are in their reproductive years.[1] While future fertility is a priority for survivors, pregnancy prevention through use of effective contraception is important for sexually active

Corresponding Author Brooke Cherven, PhD, Aflac Cancer and Blood Disorders Center, 2015 Uppergate Dr 4th Floor, Atlanta, GA 30322, bcherve@emory.edu.

Conflict of Interest All authors report no conflicts of interest

Data Sharing Data will be shared upon reasonable request to the corresponding author.

Prior Presentation of Data: Portions of this work have been presented at the International Symposium for Late Complications of Childhood Cancer (2023) and the American Psychological Association (2024).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

survivors who do not currently desire a pregnancy. Unplanned pregnancy is associated with negative social (e.g., reduced support, family/partner relationship dysfunction), psychological (e.g., depression, stress), and economic (e.g., interruption in education, lower socioeconomic status) consequences. [2–4] In addition, survivors of childhood cancer are at greater risk for maternal morbidity compared with non-survivor peers, making prevention of unplanned pregnancies a priority for this population.[5]

Compared to non-survivor peers, adult survivors are less likely to use consistent and effective contraception, more likely to use emergency contraception, and experience a similar rate of unplanned pregnancy[6–8], however less is known regarding influences on risk for unplanned pregnancy in survivors of childhood cancer. The purpose of this study was to identify sociodemographic, clinical, and psychological factors associated with risk for unplanned pregnancy among female survivors of childhood cancer to inform interventions and guide contraception counseling in this population.

Methods

Female childhood cancer survivors were recruited from 4 cancer centers in the US.[9] Eligibility criteria included ages 18–29 years, cancer diagnosis <21 years, >1 year from cancer treatment completion, and a history of gonadotoxic treatment[10]. Participants reported socio-demographics, attainment of developmental milestones (living and financial independence, full-time employment; scored on scale of 0=not true to 2=very true)[11,12], and reproductive health behaviors (current sexual activity, prior pregnancy, method(s) of contraception to prevent pregnancy, and current desire for pregnancy) through a web-based survey. Participants rated their perceived risk for infertility (none, low risk, moderate risk, high risk or unsure) and reported diagnosis of ovarian failure/premature menopause. Reproductive concerns were assessed using three subscales of the modified Reproductive Concerns Scale[13] (fertility concerns, illness concerns, information seeking; scores range 1 to 5), with higher scores indicating greater concerns. Clinical data (cancer diagnosis, age at diagnosis, treatment-related risk for infertility[10]) were abstracted from participant medical records.

Participants who reported current sexual activity (within previous 2 months) were included. Contraception method effectiveness was categorized using the World Health Organization (WHO) tiers.[14] Contraception was categorized as effective if endorsing a WHO Tier 1 method (i.e., intrauterine device, implant) or a WHO Tier II method (i.e., oral contraceptives, injectable, patch, vaginal ring). Contraception was categorized as less effective if endorsing a Tier III (i.e., condoms) or IV method (i.e., withdrawal). Those who denied any current method of contraception were categorized as no contraception. If more than one contraception method was reported, participants were categorized according to the method with the highest level of effectiveness.

Current desire for pregnancy was assessed using a five-point Likert scale (1=strongly want to avoid getting pregnant, 5=strongly want to get pregnant) which was dichotomized as *does not desire pregnancy* (responses 1–3) or *desires pregnancy* (responses 4–5). Participants were categorized based on desire for pregnancy and method of contraception. Those who

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

reported not currently desiring a pregnancy *and* using less effective/no contraception were considered at-risk for unplanned pregnancy.

Statistical Analysis

Multivariable logistic regression models were used to identify odds ratios (OR) and 95% confidence intervals (95%CI) for factors associated with risk for unplanned pregnancy. Variables with significance at p<0.1 on univariable analysis were included in a multivariable model; for variables with >5% missing data, 'Missing' was included as a category in the model. The parsimonious multivariable model was developed using stepwise backward elimination, only retaining variables with p<0.1 in the final model; variables with p<.05 were considered statistically significant. Analyses were conducted using R Studio version 2023.03.0.

The study was approved by all participating site institutional review boards and participants provided informed consent prior to enrollment.

Results

Of 325 participants, N=160 reported current sexual activity and were included in this analysis. Participants were an average of 24.0±3.1 years of age at the time of survey completion and the majority (73.8%) were non-Hispanic white (Table 1).

Over half of participants (57.5%) reported effective contraception (WHO Tier I n=40, WHO Tier II n=52), 23.8% less effective contraception (WHO Tier III n=18, WHO Tier IV n=20), and 18.8% no contraception (n=30). Most (80.6%) did not currently desire a pregnancy. Fifty-three (33.1%) participants were categorized as at-risk for unplanned pregnancy (Figure 1).

All variables in Table 1 were assessed for relation to risk for unplanned pregnancy using univariable analysis. In the parsimonious multivariable logistic regression model (Table 1), controlling for age, participants were less likely to be categorized as at-risk for unplanned pregnancy if they reported a diagnosis of ovarian failure/premature menopause (OR 0.16, 95%CI 0.02–0.72, p=.032), greater concerns about future fertility (OR 0.59, 95%CI 0.43–0.80, p=.001), and religious identity of agnostic/atheist compared with Christian (OR 0.10, 95%CI 0.01–0.41, p=.005).

Conclusion

The current findings are unique in that we identified survivors at-risk for unplanned pregnancy based on both current contraceptive use and self-reported desire for pregnancy and found that one-third of survivors were at-risk for unplanned pregnancy. The majority of sexually active survivors using less effective/no contraception reported not wanting a pregnancy, placing them at greatest risk for undesired pregnancy. Given the potential negative social, psychological, economic, and health consequences of unplanned pregnancy[2–5], strategies for incorporating comprehensive contraception counseling into cancer survivorship care are needed.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Greater fertility-related concerns were associated with reduced risk for unplanned pregnancy. In other words, those who had a stronger desire for future parenthood or worry about not being able to have a child in the future[13] were less likely to be categorized as at-risk for unplanned pregnancy. This could be a function of survivors who worry about future fertility are more engaged in reproductive healthcare and take part in discussions around contraception.[15]

Pregnancy in the setting of premature ovarian insufficiency is rare, but not impossible, therefore all survivors who do not desire a pregnancy should be counseled to use contraception. In our study, participants who reported they had previously been diagnosed with ovarian failure/premature menopause were less likely to be categorized as at risk for unplanned pregnancy. These survivors may be using oral contraceptives to manage hormonal symptoms or may incorrectly perceive their hormone replacement therapy (HRT) as contraception. Survivors with premature ovarian insufficiency receiving HRT (e.g., transdermal estradiol or oral micronized estrogen with cyclic progestins) should be counseled that HRT alone is not contraception. Compared with survivors who identified as Christian, those who were atheist or agnostic were less likely to be at-risk for unplanned pregnancy. In the general population, religious association has been related to non-use of contraception[16] and reproductive health literacy is lower among young adults with a Christian background.[17] Family background and values impact many aspects of reproductive health and counseling regarding contraception should take into account any religious considerations identified by the survivor.

There are limitations for this study. Current sexual activity was defined as vaginal or anal sex in the past two months. While sexual orientation was not associated with risk for unplanned pregnancy, it is possible that participants were engaging in sexual activity that would not place them at risk for pregnancy (e.g., vaginal sex with a female partner, anal sex only) and this could influence their contraceptive choices. Partner fertility status was not assessed and can also impact contraceptive choices. Finally, ovarian failure/premature menopause diagnoses were not confirmed via medical records, however, research has demonstrated that young adult self-report of premature ovarian insufficiency is reasonable.[18]

With one-third of survivors at-risk for unplanned pregnancy, counseling and provision of highly effective contraception is warranted during survivorship. Ongoing discussions should be tailored to the individual’s priorities for family building and any religious preferences. Finally, partnerships between survivorship clinics and gynecology, with routine annual gynecologic evaluation, are important to support survivors’ contraceptive and reproductive health needs.

Acknowledgements

Funding

Research reported in this publication was supported by the National Institute Of Nursing Research of the National Institutes of Health under Award Number K23NR020037 (Cherven, PI). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Abbreviation Key

WHO	World Health Organization
SD	Standard deviation
OR	Odds ratio
CI	Confidence interval

References

1. Howlader N NA, Krapcho M, Miller D, Brest A, Yu M, Ruhl J, Tatalovich Z, Mariotto A, Lewis DR, Chen HS, Feuer EJ, Cronin KA,. SEER Cancer Statistics Review, 1975–2018, National Cancer Institute. National Cancer Institute; 2021.
2. Nelson HD, Darney BG, Ahrens K, et al. Associations of Unintended Pregnancy With Maternal and Infant Health Outcomes: A Systematic Review and Meta-analysis. JAMA. 2022;328(17):1714–1729. doi:10.1001/jama.2022.19097 [PubMed: 36318133]
3. Barton K, Redshaw M, Quigley MA, Carson C. Unplanned pregnancy and subsequent psychological distress in partnered women: a cross-sectional study of the role of relationship quality and wider social support. BMC Pregnancy and Childbirth. 2017/01/26 2017;17(1):44. doi:10.1186/s12884-017-1223-x [PubMed: 28122585]
4. Gipson JD, Koenig MA, Hindin MJ. The effects of unintended pregnancy on infant, child, and parental health: a review of the literature. Stud Fam Plann. Mar 2008;39(1):18–38. doi:10.1111/j.1728-4465.2008.00148.x [PubMed: 18540521]
5. Zgardau A, Ray JG, Baxter NN, et al. Obstetrical and Perinatal Outcomes in Female Survivors of Childhood and Adolescent Cancer: A Population-Based Cohort Study. J Natl Cancer Inst. Apr 11 2022;114(4):553–564. doi:10.1093/jnci/djac005 [PubMed: 35043954]
6. Britton L Unintended Pregnancy: A Systematic Review of Contraception Use and Counseling in Women With Cancer. Clin J Oncol Nurs. Apr 1 2017;21(2):189–196. doi:10.1188/17.Cjon.189-196 [PubMed: 28315546]
7. Medica ACO, Stark SS, Hadnott TN, et al. Use of emergency contraception among female young adult cancer survivors. Fertility and sterility. 2018;109(6):1114–1120.e1. doi:10.1016/j.fertnstert.2018.02.136 [PubMed: 29935646]
8. Shandley LM, Kipling LM, Spencer JB, Morof D, Mertens AC, Howards PP. Factors Associated with Unplanned Pregnancy Among Cancer Survivors. J Womens Health (Larchmt). May 2022;31(5):665–674. doi:10.1089/jwh.2021.0176 [PubMed: 34860591]
9. Cherven B, Ivankova NV, Spencer JB, et al. Examining decisional needs and contextual factors influencing fertility status assessment among young female survivors of childhood cancer: A sequential mixed methods study protocol. PLoS One. 2023;18(6):e0286511. doi:10.1371/journal.pone.0286511
10. Meacham LR, Burns K, Orwig KE, Levine J. Standardizing Risk Assessment for Treatment-Related Gonadal Insufficiency and Infertility in Childhood Adolescent and Young Adult Cancer: The Pediatric Initiative Network Risk Stratification System. J Adolesc Young Adult Oncol. May 26 2020;doi:10.1089/jayao.2020.0012
11. Arnett JJ. Conceptions of the Transition to Adulthood: Perspectives From Adolescence Through Midlife. Journal of Adult Development. 2001/04/01 2001;8(2):133–143. doi:10.1023/A:1026450103225
12. Sharon T Constructing Adulthood: Markers of Adulthood and Well-Being Among Emerging Adults. Emerging Adulthood. 2016/06/01 2015;4(3):161–167. doi:10.1177/2167696815579826
13. Cherven B, Meacham L, Williamson Lewis R, Klosky JL, Marchak JG. Evaluation of the Modified Reproductive Concerns Scale Among Emerging Adult Cancer Survivors. J Adolesc Young Adult Oncol. Dec 2021;10(6):661–667. doi:10.1089/jayao.2020.0219 [PubMed: 33769891]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

14. World Health Organization Department of Sexual and Reproductive Health and Research (WHO/SRH) and Johns Hopkins Bloomberg School of Public Health/ Center for Communication Programs (CCP) KS. Family Planning: A Global Handbook for Providers WHO; 2022.
15. Cherven B, Kelling E, Lewis RW, Pruett M, Meacham L, Klosky JL. Fertility-related worry among emerging adult cancer survivors. J Assist Reprod Genet. Dec 2022;39(12):2857–2864. doi:10.1007/s10815-022-02663-1 [PubMed: 36447078]
16. Kramer MR, Hogue CJ, Gaydos LM. Noncontracepting behavior in women at risk for unintended pregnancy: what's religion got to do with it? Ann Epidemiol. May 2007;17(5):327–34. doi:10.1016/j.annepidem.2006.10.016 [PubMed: 17395484]
17. Jones SC, Boniface ER, Boehnlein C, Jones Packham A, Baldwin MK. Menstrual Health Literacy among Adolescents and Young Adults Who Menstruate: Impact of Christian Family Background and United States Region. J Pediatr Adolesc Gynecol. Dec 2023;36(6):518–524. doi:10.1016/j.jpag.2023.07.006 [PubMed: 37482080]
18. George SA, Williamson Lewis R, McKenzie L, et al. Assessment of ovarian function in adolescents and young adults after childhood cancer treatment-How accurate are young adult/parent proxy-reported outcomes? Pediatr Blood Cancer. Sep 10 2019:e27981. doi:10.1002/pbc.27981

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

		Contraception Method	
		Effective	Less Effective/None
Current Pregnancy Desire	Does not desire pregnancy	76 (47.5%)	53 (33.1%)
	Desires pregnancy	16 (10.0%)	15 (9.4%)

At risk for unintended pregnancy

Figure 1.
Contraception method by current pregnancy desire (N=160).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1.

Participant characteristics and logistic regression of factors associated with risk for unintended pregnancy

	Entire Sample	At Risk	Not at Risk	Univariable Analysis				Parsimonious Multivariable Model			
	N=160	n=53	n=107	OR	95%CI		P	OR	95%CI		P
	Mean (SD)	Mean (SD)	Mean (SD)								
Age at survey completion (years)	24.0 (3.1)	24.2 (3.1)	24.0 (3.1)	1.02	0.92	1.14	.659	1.11	0.96	1.29	.181
Age at cancer diagnosis (years)	12.4 (5.3)	12.8 (4.9)	12.1 (5.4)	1.02	0.96	1.10	.423				
	N (%)	N (%)	N (%)								
Race/Ethnicity ‡											
Non-Hispanic white	118 (73.8)	41 (77.4)	77 (72.0)	Ref							
Black	14 (8.8)	3 (5.7)	11 (10.3)	0.75	0.33	1.95	.466				
Hispanic	20 (12.5)	7 (13.2)	13 (12.1)								
Asian	1 (0.6)	0 (0.0)	1 (0.9)								
More than one-race/other	7 (4.4)	2 (3.8)	5 (4.7)								
Annual household income											
<$20,000	92 (58.6)	31 (59.6)	61 (58.1)	Ref							
$20K - $69,999	27 (17.2)	13 (25.0)	14 (13.3)	1.83	0.76	4.38	.174				
≥$70,000	38 (24.2)	30 (15.3)	8 (28.6)	0.52	0.20	1.23	.156				
Educational attainment											
High school	32 (20.0)	14 (26.4)	18 (16.8)	Ref				Ref			
Some college	54 (33.8)	19 (35.8)	35 (32.7)	0.69	0.28	1.72	.430	0.90	0.31	2.60	.839
College graduate or graduate school	74 (46.2)	20 (37.8)	54 (50.5)	0.48	0.20	1.14	**.093**	0.33	0.11	1.00	.053
Religious orientation											
Christian	91 (56.9)	34 (64.2)	57 (53.3)	Ref				Ref			
Agnostic/Atheist	26 (16.2)	2 (3.8)	24 (22.4)	0.14	0.02	0.51	**.010**	0.10	0.01	0.41	**.005**
Other (e.g., Muslim, Jewish, Buddhist)	17 (10.6)	4 (7.5)	13 (12.1)	0.52	0.14	1.59	.279	0.39	0.09	1.38	.169
Missing	26 (16.2)	13 (24.5)	13 (12.1)	1.68	0.69	4.07	.245	1.72	0.65	4.57	.273
Insurance Type											
Private	100 (64.9)	34 (66.7)	66 (64.1)	Ref							
Public	46 (29.9)	13 (25.4)	33 (32.0)	0.76	0.35	1.62	.491				

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

	Entire Sample	At Risk	Not at Risk	Univariable Analysis				Parsimonious Multivariable Model			
	N=160	n=53	n=107	OR	95%CI		P	OR	95%CI		P
	Mean (SD)	Mean (SD)	Mean (SD)								
No insurance	8 (5.2)	4 (7.8)	4 (3.9)	1.94	0.44	8.67	.369				
Relationship status											
Committed relationship	120 (75.0)	36 (67.9)	84 (78.5)	Ref							
Single	40 (25.0)	17 (32.1)	23 (21.5)	1.72	0.82	3.61	.148				
Sexual orientation and gender identity											
Cisgender and heterosexual	128 (80.0)	40 (75.5)	88 (82.2)	Ref							
LGBTQ+[†]	32 (20.0)	13 (24.5)	19 (17.8)	1.51	0.67	3.33	.315				
Cancer diagnosis category											
Leukemia	47 (29.6)	18 (33.9)	29 (27.4)	Ref							
Lymphoma	42 (26.4)	17 (32.1)	25 (23.6)	1.10	0.47	2.58	.834				
Solid tumor	49 (30.8)	13 (24.5)	36 (34.0)	0.58	0.24	1.37	.220				
Central nervous system tumor	21 (13.2)	5 (9.4)	16 (15.1)	0.50	0.14	1.54	.248				
Self-reported diagnosis of ovarian failure or early menopause											
No	144 (90.0)	51 (96.3)	93 (86.9)	Ref				Ref			
Yes	16 (10.0)	2 (3.8)	14 (13.1)	0.26	0.04	0.98	**.083**	0.16	0.02	0.72	**.032**
Treatment-related risk for future infertility											
Minimally increased risk	87 (54.7)	31 (54.8)	56 (52.8)	Ref							
Significantly increased risk	21 (13.2)	8 (15.1)	13 (12.2)	1.11	0.40	2.94	.833				
High level of increased risk	51 (32.1)	14 (26.4)	37 (34.9)	0.668	0.31	1.44	.324				
Self-perceived risk for infertility (n=158)											
Low/moderate	94 (59.5)	30 (57.7)	64 (60.4)	Ref							
High	32 (20.3)	9 (17.3)	23 (21.7)	0.83	0.33	1.97	.689				
None/unsure	32 (20.3)	13 (25.0)	19 (17.9)	1.46	0.63	3.33	.371				
Previous pregnancy											
No	119 (74.8)	37 (69.8)	82 (77.4)	Ref							
Yes	40 (25.2)	16 (30.2)	24 (22.6)	1.48	0.70	3.09	.303				
	Mean (SD)	Mean (SD)	Mean (SD)								

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

	Entire Sample	At Risk	Not at Risk	Univariable Analysis				Parsimonious Multivariable Model			
	N=160	n=53	n=107	OR	95%CI		P	OR	95%CI		P
	Mean (SD)	Mean (SD)	Mean (SD)								
Developmental milestone attainment (n=156)	1.37 (0.60)	1.32 (0.66)	1.4 (0.56)	0.81	0.46	1.42	.450				
Modified Reproductive Concerns Subscales											
Fertility concerns (n=155)	3.62 (1.30)	3.23 (1.33)	3.81 (1.24)	0.71	0.54	0.92	**.010**	0.59	0.43	0.80	**.001**
Illness concerns (n=155)	2.99 (1.24)	2.67 (1.22)	3.16 (1.22)	0.72	0.54	0.95	**.024**				
Information seeking (n=157)	3.58 (1.08)	3.54 (1.13)	3.61 (1.06)	0.94	0.69	1.28	.708				

SD = standard deviation; OR = odds ratio; CI = confidence interval

[‡]For analysis, race/ethnicity was dichotomized to compare non-Hispanic white to all other races/ethnicities, due to sample size.

[†]The n=32 survivors who identify as LGBTQ+ self-reported sexual orientation and gender identity: n=1 identified as lesbian/gay, n=19 bisexual, n=11 queer, pansexual, and/or questioning, and n=1 as something else but chose not to specify; Of these n=32, n=5 also identified as genderqueer/gender nonconforming neither exclusively male or female.